

Module 01 Guided Notes

Sound, Levels, and Engineering Reasoning

Learner:

Date started:

Date completed:

Instructions

Complete this file while studying lesson-notes.md. Write explanations in your own words. For calculations, show the formula, substitution with units, result, and plausibility check.

Use these confidence labels:

- **Green:** I can explain and apply this without help.
- **Yellow:** I understand it when I see it but cannot yet reproduce it independently.
- **Red:** I do not yet understand it.

Do not use the answer key until you have made a genuine attempt.

Checkpoint 1: Quantities and Units

Key Ideas

A physical quantity consists of a numerical value and a _____.

The equation for wavelength is:

Define each variable and its unit:

Variable	Meaning	Unit
λ		
c		
f		

Dimensional Check

Show why dividing metres per second by hertz produces metres:

Explain in Your Own Words

Why is $c = 343$ incomplete?

Quick Application

Calculate the wavelength of a 700 Hz tone using $c = 343$ m/s.

Estimate before calculating:

Formula:

Substitution with units:

Result:

Plausibility check:

Confidence: Green / Yellow / Red

Checkpoint 2: Ratios and Proportions

A ratio compares one quantity with a _____ quantity.

Complete the interpretations:

Ratio	Interpretation
4	
1	
0.25	

If a quantity follows an inverse relationship and distance doubles, the quantity becomes _____.

If a quantity follows an inverse-square relationship and distance doubles, the quantity becomes _____.

Explain in Your Own Words

What is the difference between an inverse relationship and an inverse-square relationship?

Quick Application

A power changes from 0.5 W to 2 W.

Ratio:

Meaning in words:

Confidence: Green / Yellow / Red

Checkpoint 3: Powers of Ten and Logarithms

Complete:

$$1,000 = 10^{\underline{\quad}}$$

$$0.01 = 10^{\underline{\quad}}$$

$$20 \text{ micropascals} = \underline{\hspace{2cm}} \text{ Pa}$$

A base-10 logarithm asks:

Complete:

x	$\log_{10}(x)$
0.1	
1	
10	
100	

Why is $\log_{10}(1) = 0$?

What sign does the logarithm of a ratio between zero and one have?

Quick Application

Without a calculator, predict whether each result is positive, zero, or negative:

Expression	Prediction
$\log_{10}(5)$	
$\log_{10}(1)$	
$\log_{10}(0.2)$	

Confidence: Green / Yellow / Red

Checkpoint 4: Frequency, Period, and Wavelength

Frequency describes:

Period describes:

The relationship between frequency and period is:

If frequency increases, period _____.

If sound speed stays constant and frequency increases, wavelength _____.

Calculation A

Find the period of a 50 Hz tone.

Estimate:

Formula:

Work with units:

Result in seconds:

Result in milliseconds:

Calculation B

Find the wavelength of a 2 kHz tone using $c = 343 \text{ m/s}$.

Estimate:

Formula:

Work with units:

Result:

Engineering Connection

Give one reason wavelength matters in each context:

Room acoustics:

Automotive audio:

Aerospace measurement:

Confidence: Green / Yellow / Red

Checkpoint 5: What a Decibel Means

A decibel expresses a logarithmic _____.

Why is “80 dB” incomplete?

List at least three pieces of context that can make a level statement more complete:

- 1.
- 2.
- 3.

What does 0 dB SPL mean?

Does 0 dB SPL mean there is no sound pressure? Explain.

Confidence: Green / Yellow / Red

Checkpoint 6: Power and Pressure Relationships

Write the power-ratio equation:

Write the pressure-ratio equation:

Complete:

Change	Approximate level difference
Double power	
Half power	
Double pressure	
Half pressure	

Derivation Explanation

Why does the pressure equation use 20 rather than 10?

Calculation A: Power

Power increases from 3 W to 12 W.

Identify the quantity type:

Select 10 or 20:

Ratio:

Calculation:

Result:

Plausibility check using doublings:

Calculation B: Pressure

RMS pressure decreases from 0.8 Pa to 0.2 Pa.

Identify the quantity type:

Select 10 or 20:

Ratio:

Calculation:

Result:

Plausibility check using halvings:

Confidence: Green / Yellow / Red

Checkpoint 7: Sound-Pressure Level

Write the SPL equation:

For airborne sound, the reference pressure is:

$p_0 =$ _____

Calculation

Calculate the SPL corresponding to 0.2 Pa RMS .

Formula:

Substitution:

Result:

Unit and reference statement:

Reverse Calculation

Write the equation used to convert a pressure-level difference back into a pressure ratio:

What pressure ratio corresponds to $+12 \text{ dB}$?

Estimate using known 6 dB relationships:

Calculation:

Result:

Confidence: Green / Yellow / Red

Checkpoint 8: Combining Levels

Why can two independent 70 dB sources not be added as $70 + 70 = 140 \text{ dB}$?

Write the equation for combining independent levels:

Two equal independent sources combine to approximately _____ dB above either source alone.

Calculation

Combine independent levels of 74 dB and 68 dB .

Prediction: The result must be greater than _____ and less than _____.

Calculation:

Result:

Limitation

Why might coherent signals combine differently?

Confidence: Green / Yellow / Red

Checkpoint 9: Distance and Models

Write the ideal free-field level-change equation:

In the ideal point-source model, doubling distance produces approximately _____ dB of change.

Calculation

A listener moves from 1.5 m to 6 m.

Distance ratio:

Number of doublings:

Calculated level change:

Assumptions

List four reasons a real room might not follow the ideal relationship:

- 1.
- 2.
- 3.
- 4.

Explain the difference between a useful model and a guaranteed prediction:

Confidence: Green / Yellow / Red

Checkpoint 10: Measurement Quality

Define each term in your own words:

Accuracy:

Precision:

Resolution:

Repeatability:

Can an instrument be precise but inaccurate? Explain with an example.

Why do extra decimal places not necessarily improve measurement quality?

What is calibration?

List three elements of an acoustic measurement chain:

- 1.
- 2.
- 3.

Confidence: Green / Yellow / Red

Checkpoint 11: Engineering Reasoning

For the following classroom statement, classify each sentence.

At the rear seat, the application indicated 68 dBA. The front position indicated 75 dBA. Greater source distance may contribute to the difference. The loudspeaker should be aimed and measured before replacement equipment is purchased.

Sentence	Observation, calculation, inference, or recommendation?
Rear indicated 68 dBA	
Front indicated 75 dBA	
Distance may contribute	
Aim and measure before purchase	

Write one assumption that might be required:

Write one limitation of the evidence:

Write one missing question you would ask:

Confidence: Green / Yellow / Red

Integrated Practice

Scenario

A small loudspeaker produces an indicated level of 82 dB at 2 m in a test space. A listener position is 8 m away.

Step 1: Ideal Model

Calculate the ideal free-field level at 8 m.

Known:

Unknown:

Assumptions:

Calculation:

Modeled result:

Step 2: Engineering Interpretation

Would you expect the actual room result to equal the model exactly? Why or why not?

Step 3: Recommendation

What measurement would you perform next?

Step 4: Communication

Write a three-sentence client-facing explanation that distinguishes the modeled result from a measured result.

Concept-Check Responses

Answer the ten concept checks at the end of lesson-notes.md.

1

2

3

4

5

6

7

8

9

Reflection and Readiness

Which idea feels strongest?

Which idea remains least clear?

Which calculation produced an unexpected result?

What mistake are you now less likely to make?

Can you explain why the decibel equation sometimes uses 10 and sometimes 20 without reading the lesson?

Can you identify the assumptions behind the 6 dB-per-distance-doubling model?

Readiness Decision

- ☐ Ready for practice problems
- ☐ Review one or more yellow checkpoints first
- ☐ Request another explanation or worked example before continuing